IT214 Database Management System Project Milestone 1

Lab Group 5

Section B

Group Members :

No.	Name	Student ID
1	Khushi Shah	202401471
2	Tanish Sanghavi	202401465
3	Vatsal Mori	202401436
4	Bhavya Modi	202401409

ONLINE TRIP AND TRAVEL BOOKING PLATFORM

Description of the scenario :

Planning a trip usually means jumping between multiple websites, one for flights, another for trains, a different one for hotels, and yet another for a tour package. We end up comparing prices in multiple tabs, making separate bookings, and keeping track of confirmations from different places. This platform brings all of that under one roof.

Airlines, railway operators, bus companies, hotels, and travel agencies register on the platform and manage their own listings. Each vendor is fully in control of what they list, how they price it, and what their cancellation terms are. A travel agency for instance can put up a fully planned 7-day Rajasthan package with day-by-day

itineraries, hotel stays, and transport included. A hotel can list all its room types with different pricing for peak and off-peak dates. Everything a vendor updates reflects immediately in what travellers see when they search.

Travelers browse and compare options without needing an account. Once they register, they can make bookings, apply discount codes, and manage all their trips from a single dashboard. After a trip is done, they can rate and review the service . The platform also automatically highlights services that are seeing high booking activity, so travellers know what is trending or filling up fast without anyone manually updating it.

The platform admin keeps the overall system running smoothly like approving vendors, monitoring transactions, processing refunds, and tracking how the platform is performing financially and operationally across all service types, vendors, and time periods.

1. **Objective :**

   The objective of this database is to manage all data required to run an online trip booking platform. It stores information about vendors (transport systems, hotels, tour operators) and their service listings, traveller registrations and bookings, cancellations and refunds, reviews and ratings, and promo codes. The database is also designed to support financial reporting and answer operational queries for travellers, vendors, and the platform admin.

2. **Application Users :**

- **Guest (Unregistered User)** – Can browse and search services but cannot book without signing in.
- **Traveller (Registered Customer)** – Can search, book, cancel, review, and opt for insurance policy.
- **Vendor (Service Provider)** – A registered organization (Transport Company, Hotel, or Tour Manager) that manages service listings and views sales reports.

- **Platform Admin** – Manages vendor approvals, oversees all bookings, handles refunds, and generates platform-level reports.

3. **Use cases :**

3.1. Guests

- Search for transport mode by source city, destination city, and travel date.
- Browse holiday packages by destination, duration, or price range.
- Search for hotels by city and check-in/check-out dates.
- View ratings and reviews from other travellers for any service.
- View details of any flight, hotel, or package listing.
- Compare prices across different vendors for the same route or destination.

3.2. Travellers

- Register on the platform and manage profile details.
- Search and book flights, trains and buses, hotels, or holiday packages. Provide their details and confirm booking with a promo code if applicable.
- Cancel a booking and track refund status.
- Save services to wish lists and manage them.
- Rate and review an airline, hotel, or package after a completed booking.

3.3. Vendors

- Register as an organization with company details, GST number, and vendor type (Airline / Train Operator / Bus Operator / Hotel / Tour Operator).
- Transport vendors (airline, train, bus) add routes, schedules, seat/berth classes, and manage date-wise pricing and availability.

- Hotel vendors register properties, add room types, and manage date-wise room availability.
- Tour operator vendors create holiday packages with itinerary, inclusions, departure slots, and pricing.
- Define cancellation policies for each service listing.
- View sales reports, upcoming bookings, cancellations, and customer feedback.

3.4. Platform Admin

- Approve or reject vendor registrations.
- Monitor all bookings, cancellations, and refunds across the platform.
- Generate financial reports like revenue earned, vendor payouts due, and seasonal trends.
- View ranking reports for vendors and services based on booking volume and ratings.
- View and respond to customer complaints or disputes.

4. **Key Data and Entities :**

4.1 User Registration and Identity

Travellers and Vendors are required to register on the platform before doing any activity. On registration, each gets a unique ID. A Traveller is an individual and registers by providing their name, email, phone number, date of birth, home city, and account status. A Vendor must be an organization and must have a valid GST/Tax Identification Number. For vendors, the following details are captured: Company Name, Registered Address, Website (optional), Contact Person (Name, Designation, Phone, Email), and Vendor Type. A vendor can declare one or more types at registration like Airline, Train Operator, Bus Operator, Hotel, or Tour Operator.

4.2 Service Listings

Transport Schedule covers flights, trains, and buses under one structure. Each schedule has a vendor, transport mode, route (source to destination), days of operation, departure and arrival time, and a vehicle identifier. Every schedule has seat/berth class configurations , for example flights have Economy, Business, and First Class; trains have Sleeper, 3AC, 2AC, and 1AC; buses have Seater, Semi-Sleeper, and Sleeper. Each class stores total seat count and base price. Since prices can vary by date, availability and pricing are also maintained per departure date.

Hotel Property stores name, star rating, full address, amenities as tags, check-in and check-out time, and links to the vendor. Each property has multiple room types, each with a name, guest capacity, price per night, and total room count. Since availability changes daily, the number of available rooms per room type is tracked date-wise.

Holiday Package stores name, theme, destinations, duration, inclusions, price per person, and links to the tour operator vendor. A package is offered on specific departure dates, each departure slot stores the date, available seat count, and price for that slot.

4.3 Booking and Transactions

Booking is the central transaction entity. It stores a unique booking ID, booking date, type (Transport, Hotel, or Package), status (CONFIRMED, CANCELLED, or COMPLETED), total amount charged, discount applied if any, and the payment service name and reference number returned by the external gateway. A booking links one traveller to one service. Transport bookings additionally store per-passenger details such as name, age, gender, and government ID. Hotel bookings store guest names and the selected room type with dates. Package bookings store per-traveller details and the selected departure slot.

4.4 Dynamic Demand Tagging

The platform automatically computes a demand tag for every listed service based on booking frequency over a rolling time window. No manual input is required from vendors. The database stores the tag value (HIGH DEMAND, FILLING FAST, or LOW AVAILABILITY), the date and time it was last computed. For example, if a particular train route sees a spike in bookings in the last 48 hours relative to its available seats, it gets tagged as FILLING FAST. These tags are visible to travellers while browsing and are updated periodically, helping travellers make quicker decisions and giving the platform a real-time feel .

4.5 Travel Insurance

At the time of confirming a booking, a traveller has the option to add travel insurance to their trip. The platform ties up with external insurance providers, so the database stores the Insurance Provider Name, Plan Name, Plan Type (Single Trip or Multi Trip), coverage details (such as trip cancellation cover, medical emergency cover, and loss of baggage cover), premium amount charged, and the validity period of the policy. Each insurance record is linked to a specific booking and a specific traveller. If a booking is cancelled, the insurance record is also updated to reflect whether the insurance claim was filed, and its current claim status (NOT CLAIMED, CLAIM FILED, or SETTLED).

4.6 Cancellation, Refund, and Feedback

Cancellation and Refund records the cancellation date, reason, refund amount calculated based on the vendor's cancellation policy, current refund status (PENDING, PROCESSED, or REJECTED), and the date the refund was processed.

Promo Code stores the code, discount type (flat or percentage), discount value, minimum booking amount required, validity dates, the service type it applies to, maximum allowed uses, and how many times it has been used so far.

Review and Rating stores the traveller who left the review, the type of entity being reviewed (transport provider, hotel, or package), the entity ID, a star rating from 1

to 5, a written comment, and the review date.

5. **Tentative list of Queries**

For Guests

1. Search for available transport by one or more parameters such as Source City, Destination City, Date, Transport Mode, Class.
2. Search for hotels by one or more parameters such as City, Check-in Date, Check-out Date, Star Rating, Price Range.
3. Browse holiday packages by one or more parameters such as Destination, Theme, Duration, Price Range.
4. Compare prices for the same route or destination across different vendors and transport modes along with average traveller ratings of each vendor.

For Travelers

1. My Bookings - all bookings with type, service details, travel date, and current status.
2. Upcoming Trips - confirmed bookings where travel date is in the future.
3. My Cancellations and Refund Status - cancelled bookings with cancellation date, refund amount, and whether refund is pending, processed, or rejected.
4. My Insurance - insurance policies linked to each booking with plan details and claim status.
5. My Spending Summary - total amount spent over a selected time period broken down by service type (transport, hotel, package).

For Vendors

1. Daily, Weekly, Monthly, and Annual sales and revenue summaries on various combinations of the following parameters:

- Overall sales
- Service-type wise (transport, hotel, package)
- Destination wise and route wise
- Class wise (Economy, Sleeper, Deluxe room, etc.)

2. Sales Register - date-wise log of all bookings made for the vendor's services.
3. List of upcoming bookings to be serviced
4. Cancellation summary - total cancellations, reasons, and total refund amount owed for a selected period.
5. Payment received from the platform.
6. Customers who have booked from this vendor
7. All bookings made by a specific customer for this vendor's services.
8. Customers who gave a poor rating (1 or 2 stars) along with their review comments.
9. Services currently tagged as HIGH DEMAND, FILLING FAST, or LOW AVAILABILITY based on recent booking activity.

For Platform Admin

1. Daily, Weekly, Monthly, and Annual revenue summaries on various combinations of the following parameters:
   - Overall platform revenue
   - Vendor wise, service-type wise, destination wise, and transport mode wise
2. Financial ledger of a vendor.
3. Pending refunds list.
4. Top vendors ranked by booking volume, revenue, and average customer rating.
5. Top destinations and routes by booking volume over a selected time period.

6. Customer feedback summary - all reviews for a specific vendor or a specific service with average rating.
7. Services with consistently poor ratings - average rating below 2.5 across all reviews.
8. Monthly booking volume by service type to identify peak and off-peak seasons.
9. Insurance claim summary — all bookings where a claim was filed with current status and settlement amount.